JEE(MAIN) : ENTHUSIAST SCORE-II

Time : 3.00 Hours

Q2

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet and OMR sheet (Side-1 & Side-2) with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of 90 questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics**, **Chemistry** and **Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
   (a) Section-I contains 20 **multiple choice** questions with **only one correct** option.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
   (b) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray marks on the Answer Sheet.**
9. **Take g = 10 m/s^2 unless otherwise stated.**

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Topic : Full Syllabus

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : –1 If wrong option is selected.

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1. The resistance of a moving coil galvanometer is 20Ω. It requires 0.01A current for full scale deflection. Calculate the value of resistance required to convert it into a voltmeter of range 20 volt :-

(A) 1980 Ω (B) 2000 Ω

(C) 1880 Ω (D) 1500 Ω

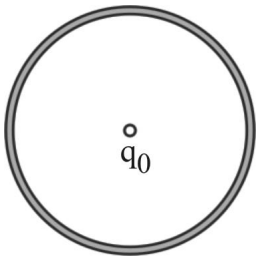
2. Two long parallel wires carrying equal currents in opposite directions are placed at x = ± a parallel to y-axis with z = 0. Magnetic field at origin O is B_1 and at P (2a, 0, 0) is B_2. Then B_1/B_2 is :-

(A) 3 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) 2

3. A series R–L–C circuit consists of an 8.0 Ω resistor, a 5.0 μF capacitor and a 50.0 mH inductor. A variable frequency source applies an emf of 400 V (rms) across the combination. The power delivered to the circuit when the frequency is equal to one-half the resonance frequency is :-

(A) 52 W (B) 57 W (C) 63 W (D) 69 W

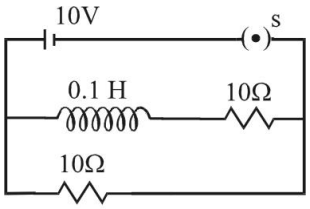
4. A ring of radius R with a uniformly distributed charge q as shown in figure. A change q_0 is now placed at the centre of the ring. Find the increment in the tension in ring :-



(A) $\frac{kqq_0}{2\pi R^2}$ (B) $\frac{kaq_0}{R^2}$

(C) $\frac{kqq_0}{\pi R^2}$ (D) None of these

5. In the adjoining circuit, initially the switch S is open. The switch 'S' is closed at t = 0. The difference between the maximum and minimum current that can flow in the circuit through battery is : –



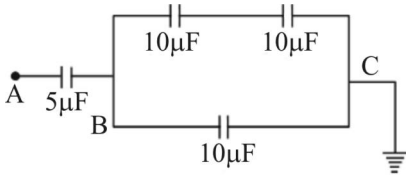
(A) 2 Amp

(B) 3 Amp

(C) 1 Amp

(D) Nothing can be concluded

6. In the given circuit if point C is connected to the earth and a potential of + 2000 V is given to point A, then potential at B is :



(A) 1500 V (B) 1000 V

(C) 500 V (D) 400 V

7. 5.6 liter of helium gas at STP is adiabatically compressed to 0.7 liter. Taking the initial temperature to be T_1, the work done in the process is :

(A) $\frac{9}{8}RT_1$ (B) $\frac{3}{2}RT_1$

(C) $\frac{15}{8}RT_1$ (D) $\frac{9}{2}RT_1$

8. A vessel contains a mixture of 1 mole of oxygen and two moles of nitrogen at 300 K. The ratio of the rotational kinetic energy per O_2 molecule to that per N_2 molecule is :

(A) 1 : 1

(B) 1 : 2

(C) 2 : 1

(D) depends on the moment of inertia of the two molecules

9. In a Young's double slit experiment the intensity at a point where the path different is $\frac{\lambda}{6}$ (λ being the wavelength of the light used) is I. If I_0 denotes the maximum intensity, I/I_0 is equal to :-

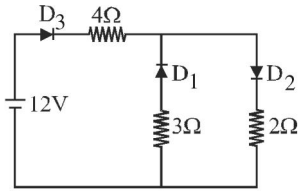
(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

10. An object is placed at a distance of 15 cm from a convex lens of focal length 10 cm. On the other side of the lens, an convex mirror is placed at its focus such that the image formed by the combination coincides with the object itself. The radius of the convex mirror is

(A) 20 cm (B) 10 cm

(C) 15 cm (D) 30 cm

11. In a circuit ideal diodes are given, find current drawn from battery?



(A) 2.00 ampere (B) 2.31 ampere

(C) 1.33 ampere (D) 1.71 ampere

12. A star initially has 10^{40} deuterons. It produces nucleus via the processes

$${}_1H^2 + {}_1H^2 \longrightarrow {}_1H^3 + p$$

and ${}_1H^2 + {}_1H^3 \longrightarrow {}_2H^4 + n$

where the masses of the nuclei are :

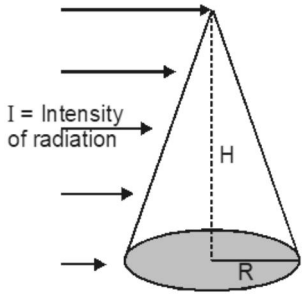
$m({}_1H^2) = 2.014$ amu, $m(p) = 1.007$ amu,

$m(n) = 1.008$ amu and $m({}_2He^4) = 4.001$ amu. If the average power radiated by the star is 10^{16} watt, the deuteron supply of the star is exhausted in a time of the order of :

(A) 10^6 s (B) 10^8 s

(C) 10^{12}s (D) None

13. The radiation force experienced by body exposed to radiation of intensity I, assuming surface of body to be perfectly absorbing is :



(A) $\frac{\pi R^2 I}{c}$ (B) $\frac{\pi RHI}{c}$

(C) $\frac{IRH}{2c}$ (D) $\frac{IRH}{c}$

14. The electric potential between a proton and an electron is given by $V = V_0 \ell n \frac{r}{r_0}$, where r_0 is a constant. Assuming Bohr's model to be applicable, write variation of r_n with n, n being the principal quantum number :-

(A) $r_n \propto n$ (B) $r_n \propto \frac{1}{n}$

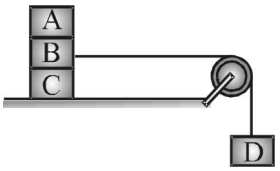
(C) $r_n \propto n^2$ (D) $r_n \propto \frac{1}{n^2}$

15. The velocity V of waves produced in water depends on their wavelength λ, the density of water ρ, and acceleration due to gravity g. The square of velocity is proportional to :-

(A) $\lambda^{-1} g^{-1} \rho^{-1}$ (B) λg

(C) $\lambda \rho g$ (D) $\lambda^2 g^{-2} \rho^{-1}$

16. These blocks A, B and C of equal mass m are placed one over the other on a smooth horizontal ground as shown in figure. Coefficient of friction between any two blocks of A, B and C is 1/2. The maximum value of mass of block D so that the blocks A, B and C move without slipping over each other is :-



(A) 6m (B) 5m

(C) 3m (D) 4m

17. A particle moving along x-axis has acceleration f at time t, given by : $f = f_0\left(1 - \frac{t}{T}\right)$, where f_0 and T are constants. The particle at t = 0 has zero velocity. In the time interval between t = 0 and the instant when f = 0, the particle's velocity (n_x) is :–

(A) $1/2\ f_0T^2$

(B) f_0T^2 (initial velocity is zero)

(C) $1/2\ f_0T$

(D) f_0T

18. A small disc of radius 2 cm is cut from a disc of radius 6 cm. If the distance between their centres is 3.2 cm, what is the shift in the centre of mass of the disc?

(A) 0.4 cm (B) 2.4 cm

(C) 1.8 cm (D) 1.2 cm

19. A ball of mass m moving horizontally at a speed u collides with the bob of a simple pendulum at rest. The mass of the bob is $\frac{m}{2}$. If the collision is perfectly inelastic the height to which the composite ball rise after the collision is :-

(A) $\frac{u^2}{2g}$ (B) $\frac{u^2}{8g}$

(C) $\frac{2u^2}{9g}$ (D) $\frac{u^2}{16g}$

20. A solid cylinder starts rolling from a height h on an inclined plane. At same instant t, the ratio of its rotational K.E. and the total K.E. would be :-

(A) 1 : 2 (B) 1 : 3

(C) 2 : 3 (D) 1 : 1

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

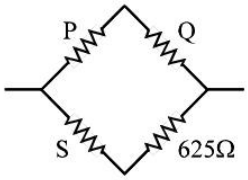
Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

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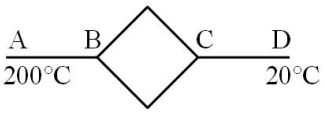
1. A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51Ω to secure balance. The unknown resistance in the fourth arm is 6.5×10^N Ω, then N will be :-



2. Two short electric dipoles of moment P and 64 P are placed in opposite direction on a line at a distance of 25 cm. The electric field will be zero at point between the dipoles whose distance from the dipole of moment P is (in cm) –

3. 2 kg of ice at – 20°C is mixed with 5 kg of water at 20°C in an insulating vessel having a negligible heat capacity. Calculate the final mass of water (In Kg) remaining in the container. It is given that the specific heats of water and ice are 1 kcal/kg per °C and 0.5 kcal/kg/°C while the latent heat of fusion of ice is 80 kcal/kg :

4. Six identical rods are arranged as shown in the figure. The temperature of the junction B (In °C) will be :-

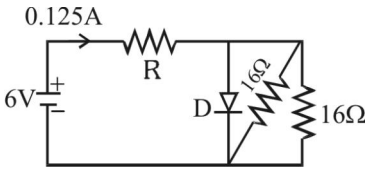


5. A convex lens of focal length 20 cm and a concave lens of focal length f are mounted coaxially 5 cm apart. Parallel beam of light incident on the convex lens emerges from the concave lens as a parallel beam. Then magnitude of 'f' in cm is

6. A glass tube of length 1.0m is completely filled with water. A vibrating tuning fork of frequency 500 Hz is kept over the mouth of the tube and the water is drained out slowly at the bottom of the tube. If velocity of sound in air is 330 ms^{-1}, then the total number of resonances that occur will be :-

7. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, $\frac{V}{4}$ the stopping potential is The threshold wavelength for the metallic surface is :-

8. In the given circuit, the diode has a forward resistance of 8 Ω and infinite backward resistance. The value of R is :-



9. The angular velocity of a body is $\vec{\omega} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and a torque $\vec{\tau} = \hat{i} + 2\hat{j} + 3\hat{k}$ acts on it. The rotational power (in watt) will be :-

10. A simple pendulum of length 1 m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1 rad/s and an amplitude of 10^{-2} m. The relative change in the angular frequency of the pendulum is $\frac{n}{1000}$. Find value of n.

Topic : Full Syllabus

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : –1 If wrong option is selected.

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1. Which represents correct order of acidic strength ?

(A) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

(B) $La(OH)_3 > Nd(OH)_3 > Lu\ (OH)_3$

(C) $NaOH < KOH < RbOH < CsOH$

(D) $CH_4 < H_2S < HCl < HI$

2. **Statement**-1 : Pb^{4+} compounds are stronger oxidizing agents than Sn^{4+} compounds.
Statement-2 : The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to inert pair effect.

(A) Statement-1 is true, Statement-2 is True, Statement-2 is a correct explanation for statement-1

(B) Statement-1 is true, Statement-2 is True, Statement-2 is not a correct explanation for statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

3. Which element is having lowest melting and boiling point?

(A) Ti (B) Cu (C) Zn (D) Mn

4. Which of the following is not formed when H_2S reacts with acidic $K_2Cr_2O_7$ solution ?

(A) K_2SO_4 (B) $CrSO_4$

(C) S (D) $Cr_2(SO_4)_3$

5. The number of unpaired electrons calculated in $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are :-

(A) 4 and 4 (B) 0 and 2

(C) 2 and 4 (D) 0 and 4

6. Which of the following complex shows optical activity ?

(A) trans-$[Co(ox)_2Cl_2]^{3-}$

(B) $[Co(NH_3)_3Cl_3]$

(C) $[Fe(edta)]^-$

(D) cis-$[Pt(NH_3)_2Cl_2]$

7. A diatomic molecule has dipole moment of 1.2 D. If its bond length is equal to 10^{-10} m then the fraction of an electronic charge on each atom will be :-

(A) 42% (B) 52%

(C) 37% (D) 25%

8. (1) $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$; K_1
(2) $CH_{4(g)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + 3H_{2(g)}$; K_2
(3) $CH_{4(g)} + 2H_2O_{(g)} \rightleftharpoons CO_{2(g)} + 4H_{2(g)}$; K_3
If the relation between K_1, K_2 and K_3 is
$K_3 = \frac{(K_1)^x}{(K_2)^y}$ then calculate $(x - y)$

(A) 2 (B) 3 (C) 1 (D) 0.5

9. Half life of a first order reaction :
$A(aq.) \longrightarrow 2B(aq.)$ is 40 min at 27°C and 20 min at 37°C. The activation energy of reaction is
($ln_2 = 0.7$, $R = 2.0$ cal $mol^{-1}K^{-1}$)

(A) 19530 cal/ mol (B) 13020 cal/ mol

(C) 26040 cal/ mol (D) 6510 cal/ mol

10. Volume of water which must be added to 8M NaOH to make 2 lit. of 20 % w/v NaOH solution is :

(A) 1.25 lit. (B) 0.75 lit.

(C) 1 lit. (D) None of these

11. Given, limiting equivalent conductance of
$BaCl_2 = x_1$, $H_2SO_4 = x_2$, $HCl = x_3$
Specific conductance of the saturated $BaSO_4$ solution = y Scm^{-1}. Thus K_{SP} of $BaSO_4$ is –

(A) $10^3y / 2(x_1 + x_2 - 2x_3)$

(B) $106y^2 / (x_1 + x_2 - 2x_3)^2$

(C) $106y^2 / 4(x_1 + x_2 - x_3)^2$

(D) $(x_1 + x^2 - 2x^3) / 106y^2$

12. Two solutions A and B, each of 100 L was made by dissolving 4g of NaOH and 9.8 g of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing 40 L of solution A and 10 L of solution B is_____.

(A) 10 (B) 11 (C) 7 (D) 12

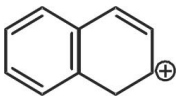
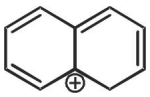
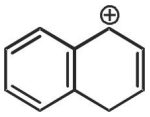
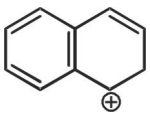
13. A hydrate of magnesium iodide has a formula $MgI_2.xH_2O$. If 1.055 g sample is heated to a constant weight of 0.695 g. What is the value of x ?

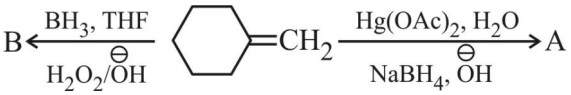
(A) 4 (B) 2 (C) 6 (D) 8

14. What is the final temperature (in kelvin) of 0.10 mole monoatomic ideal gas that performs 75 cal of work adiabatically if the initial temperature is 227ºC ? (use R = 2 cal/K-mol)

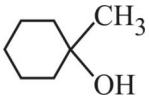
(A) 250 (B) 500 (C) 200 (D) 400

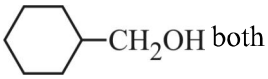
15. Among the following most stable resonating structure is :-

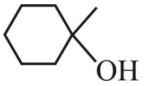
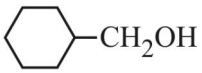
(A) 
(B) 
(C) 
(D) 

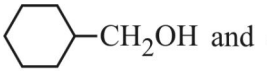
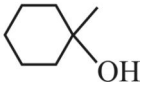
16. 

A and B are respectively :-

(A)  both

(B)  both

(C)  and 

(D)  and 

17. Match the vitamins given in Column-I with the deficiency diseases listed in Column-II

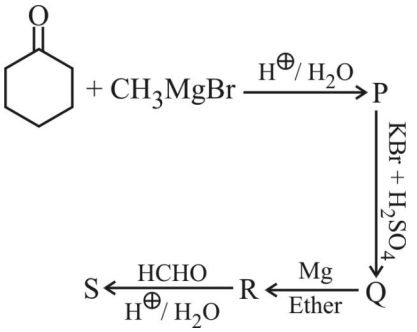
Column-I		Column-II	
(a)	Vitamin A	(p)	Osteomalacia
(b)	Vitamin C	(q)	Beri beri
(c)	Vitamin D	(r)	Scurvy
(d)	Vitamin B_1	(s)	Xerophthalmia

(A) a ⟶ p; b ⟶ q; c ⟶ s; d ⟶ r

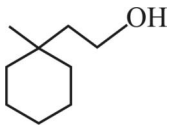
(B) a ⟶ s; b ⟶ r; c ⟶ p; d ⟶ q

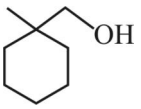
(C) a ⟶ s; b ⟶ p; c ⟶ r; d ⟶ q

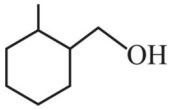
(D) a ⟶ p; b ⟶ r; c ⟶ q; d ⟶ s

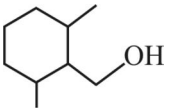
18. 

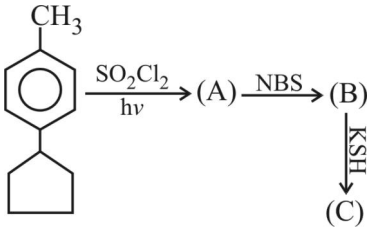
S is:

(A) 

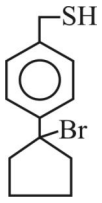
(B) 

(C) 

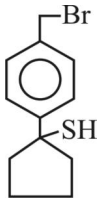
(D) 

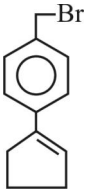
19. 

Product (C) is:

(A) 

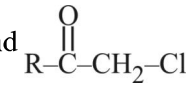
(B) 

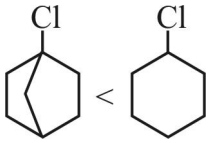
(C) 

(D) 

20. Which of the following statement is not correct ?

(A) Compound $Ph-CH_2-Cl$ can undergo S_N1 & S_N2 reactions

(B) For compound  Rate of S_N2 < Rate of S_N1

(C) Rate of < S_N1 

(D) Compound CH_3-Cl will prefer to undergo S_N2 when reacting with aq. KOH

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

---

1. The value of $(n \times \ell \times m)$ for the last electron entered in K is

2. How many of these molecules are non-polar ?

O_3, SO_3, PCl_5, CCl_4, SF_4, H_2O, PH_3, PCl_3F_2, XeF_4

3. How many of following orders are correct?

(a) Order of IE : $O^{-2} > O^+ > O > O^-$

(b) Order of EN : Zn < Cd < Hg

(c) Order of IE : B > Tl > Ga > Al > In

(d) Order of EA : S > Se > Te > O

(e) Order of atomic radius : Ni < Cu < Zn

(f) Order of ionic radius : $Na^+ > Mg^{+2} > Li^+ > Be^{+2}$

(g) Order of IE_2 : N < O < F < Ne

4. The half-life of a first order reaction is one hour. In what time will the concentration be reduced to $\frac{7}{8}$ of its initial value :- (log 7 = 0.84)

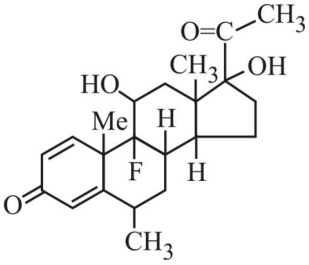
5. For HCl solution at 25°C, Equivalent conductance at infinity dilution is 425 ohm^{-1} cm^2 equivalent. The specific conductance of solution of HCl is 3.825 ohm^{-1} cm^{-1}. If the degree of dissociation is 90% the normality of the solution is :-

6. The solubility product of $Cr(OH)_3$ at 298 K is 6.0×10^{-31}. The concentration of hydroxide ions in a saturated solution of $Cr(OH)_3$ will be $(x \times 10^{-31})^{1/4}$, then x is :

7. If x gram of non-volatile, non-electrolytic solute is dissolved in 114 g octane to reduce its vapour pressure to 80% of that of pure octane then x is ____

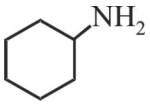
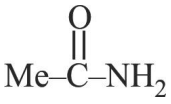
(Molar mass of solute = 40 g/mole)

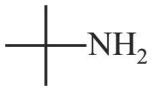
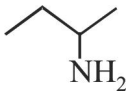
8. How many chiral centres are present in

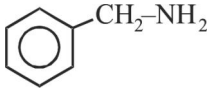


Fluoromethanolone

9. How many of the following give positive isocyanide test

(a)  (b)  (c) $Ph-NH_2$

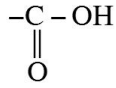
(d)  (e)  (f) 

(g)  (h) 

10. How many species is/are −M/−R groups ?

(a) $-OH$ (b) $-NO_2$ (c) $-CN$

(d) $-CH_3$ (e) $-O^{\ominus}$ (f) $-\overset{\oplus}{N}H_3$

(g)  (h) $-CH=NH$

Topic : Full Syllabus

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

---

1. The integral

$$\int \frac{(x^8 - x^2)dx}{(x^{12} + 3x^6 + 1)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$$ is equal to :

(A) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/3} + C$

(B) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/2} + C$

(C) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right) + C$

(D) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{3} + C$

2. The area enclosed by the curves $xy + 4y = 16$ and $x + y = 6$ is equal to :

(A) $28 - 30\log_e 2$ (B) $30 - 28\log_e 2$

(C) $30 - 32\log_e 2$ (D) $32 - 30\log_e 2$

3. Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $\left|P^{-1}AP - 2I\right|$ is equal to

(A) 26 (B) 27

(C) 66 (D) 23

4. An integer is chosen at random from the integers 1, 2, 3…50. The probability that the chosen integer is a multiple of atleast one of 4, 6 and 7 is :

(A) $\frac{8}{25}$ (B) $\frac{21}{50}$

(C) $\frac{9}{50}$ (D) $\frac{14}{25}$

5. If the system of linear equations

$x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

has infinitely many solutions, then $12\alpha + 13\beta$ is equal to

(A) 60 (B) 64 (C) 54 (D) 58

6. If $\log_e a$, $\log_e b$, $\log_e c$ are in an A.P. and $\log_e a - \log_e 2b$, $\log_e 2b - \log_e 3c$, $\log_e 3c - \log_e a$ are also in an A.P, then a : b : c is equal to

(A) 9 : 6 : 4 (B) 16 : 4 : 1

(C) 25 : 10 : 4 (D) 6 : 3 : 2

7. Let Ajay will not appear in JEE exam with probability $p = \frac{2}{7}$, while both Ajay and Vijay will appear in the exam with probability $q = \frac{1}{5}$. Then the probability, that Ajay will appear in the exam and Vijay will not appear is :

(A) $\frac{9}{35}$ (B) $\frac{18}{35}$

(C) $\frac{24}{35}$ (D) $\frac{3}{35}$

8. If $A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $C = ABA^T$ and $X = A^TC^2A$, then det X is equal to :

(A) 243 (B) 729

(C) 27 (D) 891

9. If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then

(A) $2S_{12} = S_{11} + S_{10}$

(B) $S_{12} = S_{11} + S_{10}$

(C) $2S_{11} = S_{12} + S_{10}$

(D) $S_{11} = S_{10} + S_{12}$

10. Function $f(x) = \frac{\lambda \sin x + 3\cos x}{2\sin x + 6\cos x}$ is monotonic increasing when :-

(A) $\lambda < 1$ (B) $\lambda > 1$

(C) $\lambda < 2$ (D) $\lambda > 2$

11. If

$$f(x) = \begin{cases} \frac{1-\sin x}{(\pi-2x)^2} \cdot \frac{\log \sin x}{\log(1+\pi^2-4\pi x+4x^2)} & , \ x \neq \frac{\pi}{2} \\ k & , \ x = \frac{\pi}{2} \end{cases}$$

is continuous at $x = \pi/2$, then k is -

(A) $-\frac{1}{16}$ (B) $-\frac{1}{32}$

(C) $-\frac{1}{64}$ (D) $-\frac{1}{128}$

12. A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is :

(A) $\frac{1}{\sqrt{7}}$ (B) $2\sqrt{2}-1$

(C) $\sqrt{7}-1$ (D) $\frac{1}{2\sqrt{2}}$

13. If $x \in (\pi, 2\pi)$ &

$\frac{\sqrt{1+\cos x}+\sqrt{1-\cos x}}{\sqrt{1+\cos x}-\sqrt{1-\cos x}} = \cot\left(a+\frac{x}{2}\right)$ then 'a' equals :-

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$ (D) None of these

14. If $\sum_{i=1}^{5}(x_i - 10) = 5$ and $\sum_{i=1}^{5}(x_i - 10)^2 = 25$, then standard deviation of observations $2x_1 + 7$, $2x_2 + 7$, $2x_3 + 7$, $2x_4 + 7$ and $2x_5 + 7$ is equal to-

(A) 8 (B) 16

(C) 4 (D) 2

15. $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to

(A) 3 (B) 2

(C) 4 (D) 1

16. Vector $\hat{a}$ in the plane of $\vec{b} = 2\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$ is such that it is equally inclined to $\vec{b}$ and $\vec{d}$ where $\vec{d} = \hat{j} + 2\hat{k}$. The value of $\hat{a}$ is :-

(A) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$ (B) $\frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$

(C) $\frac{2\hat{i}+\hat{j}}{\sqrt{5}}$ (D) None of these

17. Given A(1,2) & B(3,7) are two vertices of the ΔABC. If the locus of centroid of ΔABC is $2x - y = 0$ then the minimum distance between the locus of the vertex C from the line $2x - y = 0$ is-

(A) 0 (B) 1

(C) $\frac{2}{\sqrt{5}}$ (D) $\frac{1}{\sqrt{5}}$

18. On portion of straight line $x + y = 2$ which is intercepted between the axes, a square away from the origin is constructed. p deotes the perpendicular distance of a side of square from origin, then maximum value of p is :

(A) $\sqrt{2}$ (B) $2\sqrt{2}$ (C) $3\sqrt{2}$ (D) $4\sqrt{2}$

19. The co-ordinates of a point on the parabola $y^2 = 8x$ whose focal distance is 4 is :

(A) $(2, \pm 4)$ (B) $(\pm 2, 4)$

(C) $(-2, \pm 4)$ (D) $(\pm 2, -4)$

20. The straight lines whose direction cosines are given by $a\ell + bm + cn = 0$, $fmn + gn\ell + h\ell m = 0$ are perpendicular, if :-

(A) $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$

(B) $\sqrt{\frac{a}{f}} + \sqrt{\frac{b}{g}} + \sqrt{\frac{c}{h}} = 0$

(C) $\sqrt{af} = \sqrt{bg} = \sqrt{ch}$

(D) $\sqrt{\frac{a}{f}} = \sqrt{\frac{b}{g}} = \sqrt{\frac{c}{h}}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

---

1. If the solution curve, of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passing through the point (2, 1) is

$\tan^{-1}\left(\frac{y-1}{x-1}\right) - \frac{1}{\beta}\log_e\left(\alpha + \left(\frac{y-1}{x-1}\right)^2\right)$

$= \log_e|x-1|$, then $5\beta + \alpha$ is equal to

2. If $x = x(t)$ is the solution of the differential equation $(t + 1)dx = (2x + (t + 1)^4)\, dt$, $x(0) = 2$, then, $x(1)$ equals _____.

3. If α satisfies the equation $x^2 + x + 1 = 0$ and $(1 + \alpha)^7 = A + B\alpha + C\alpha^2$, $A, B, C \geq 0$, then $5(3A - 2B - C)$ is equal to ____.

4. In the expansion of

$$(1 + x)\left(1 - x^2\right)\left(1 + \frac{3}{x} + \frac{3}{x^2} + \frac{1}{x^3}\right)^5,$$

$x \neq 0$, the sum of the coefficient of x^3 and x^{-13} is equal to ___

5. The points of extremum of the function

$$F(x) = \int_1^x e^{-t^2/2}(1 - t^2)\, dt \text{ are :-}$$

6. If $f(x)$ is a polynomial satisfying $f(x)f(1/x) = f(x) + f(1/x)$ and $f(2) > 1$ then $\lim\limits_{x \to 1} f(x)$ is :-

7. If $\sin^2\alpha + \sin^4\alpha = 1$ & $\cos^8\alpha - \sin^8\alpha = \left(b - 9\sqrt{a}\right)$, then $\frac{b}{a}$ is equal to

8. Total number of equivalence relations defined in the set $S = \{a, b, c\}$ is

9. If the shortest distance between the line $\vec{r} = \left(-\hat{i} + 3\hat{k}\right) + \lambda(\hat{i} - a\hat{j})$ and $\vec{r} = \left(-\hat{j} + 2\hat{k}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to

10. If the area of an equilateral triangle inscribed in the circle, $x^2 + y^2 + 10x + 12y + c = 0$ is $27\sqrt{3}$ sq. units then c is equal to

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Main)
SCORE - II
20-02-2024

JEE(Main) : ENTHUSIAST COURSE SCORE -II

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	B	A	C	C	A	A	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	D	A	B	C	C	A	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	650	5	6	140	15	3	3	44	20	1

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	C	B	D	C	D	A	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	D	A	A	C	B	B	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0	5	4	12	10	18	10	8	5	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	B	D	A	B	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	A	C	A	C	D	C	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	11	14	5	118	1	2	4	5	2	25

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

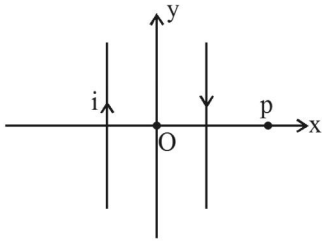
1. Ans (A)

$V = i(R + S)$

$20 = (0.01)(20 + S)$

$S = 1980\ \Omega$

2. Ans (A)



$B_1 = \frac{\mu_0 i}{2\pi a} + \frac{\mu_0 i}{2\pi a}$

$B_2 = \frac{\mu_0 i}{2\pi a} - \frac{\mu_0 i}{2\pi(3a)}$

3. Ans (B)

$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \times 10^{-3} \times 5 \times 10^{-6}}} = 2000$

$\omega = \frac{\omega_r}{2} = 1000$

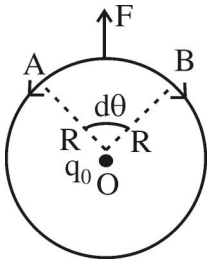
$X_L = \omega_L = 1000 \times 5 \times 10^{-3} = 50\Omega$

$X_C = \frac{1}{\omega_C} = \frac{1}{1000 \times 5 \times 10^{-6}} = 200\ \Omega$

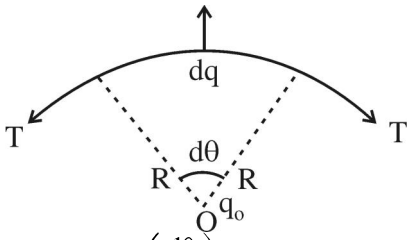
$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$P = \left(\frac{V}{Z}\right)^2 R$

4. Ans (A)



Take small segment AB



$F = 2T \sin\left(\frac{d\theta}{2}\right)$

$\frac{kq_0dq}{R^2} = 2T\left(\frac{d\theta}{2}\right)$ [$\because$ $d\theta$ is very small]

$\frac{kq_0qd\theta}{2\pi R^2} = T\, d\theta$

$T = \frac{kq_0q}{2\pi R^2}$

$\left[dq = \frac{qR}{2\pi R}d\theta \quad dq = \frac{qd\theta}{2\pi}\right]$

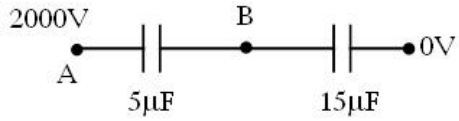
5. Ans (C)

$t = 0, i = \frac{10}{10} = 1A$

$t = \infty, i = \frac{10}{5} = 2A$

$\Delta i = 2 - 1 = 1A$

6. Ans (C)



Using voltage division

$V_B = \left(\frac{5}{20}\right) \times 200$

$V_B = 500$ V

7. Ans (A)

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$T_1(5.6)^{5/3-1} = T_2(0.7)^{5/3-1}$

$T_2 = 4T_1$

$n = \frac{5.6}{22.4} = \frac{1}{4}$

$\omega = \frac{nR\Delta t}{1-\gamma} = \frac{\frac{1}{4} \times R \times 3T_1}{1-\frac{5}{3}} = \frac{9}{8}RT_1$

8. Ans (A)

$E = \frac{NfKT}{2}$

$\frac{E}{N} = \frac{fKT}{2}$

Both are diatomic gas so ratio of energy is 1 : 1

9. Ans (D)

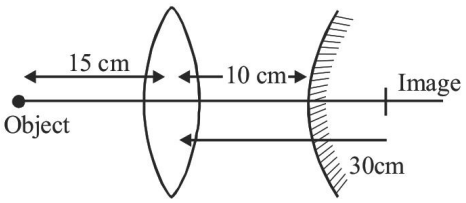
$I_0 = 4I$

$I_{resu.} = I + I + 2\sqrt{I\,I}\cos\frac{2\pi}{\lambda} \cdot \frac{\lambda}{6}$

$= 3I$

$\frac{I_{resu.}}{I_0} = \frac{3}{4}$

10. Ans (A)



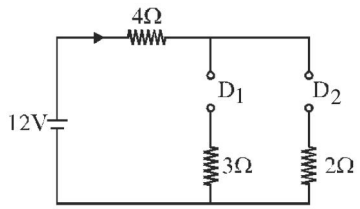
$\frac{1}{10} = \frac{1}{v} - \frac{1}{(-15)}$ R = 20 cm

v = 30 cm

11. Ans (A)

D_1 diode is in RB

but D_2 and D_3 are in FB



Apply kirchoff law,

$-12 + 4i + 2i = 0$

or $i = \frac{12}{6} = 2$ A

12. Ans (C)

$3\ {}_1H^2 \longrightarrow {}_2He^4 + P + n$

mass defect $\Delta m = 0.026$ amu

and energy per reaction $= 0.026 \times 931$

$= 24.206$ MeV

Net power $= 10^{16}$ J/s

$\therefore$ No. of deutrons conserved per second

$$= \frac{10^{16}}{3.87 \times 10^{-12}} = 2.58 \times 10^{27}/s$$

Hence, the supply of deutron will exhaust in times.

$$= \frac{10^{40}}{2.58 \times 10^{27}} = 3.87 \times 10^{12} s$$

13. Ans (D)

Radiation force

$$= \frac{I}{C} \times \text{Area} = \frac{I}{C}.\frac{1}{2}.2R\,.\,H = \frac{IRH}{C}$$

14. Ans (A)

Potential energy $U = eV = eV_0\ \ell n\left|\frac{r}{r_0}\right|$

$\therefore$ Force $F = \left|-\frac{du}{dr}\right| = \frac{eV_0}{r}$

Using Bohr's model

$$\frac{mv^2}{r} = \frac{eV_0}{r} \qquad .....(1)$$

and $mvr = \frac{nh}{2\pi}$ .....(2)

we have $r \propto n$

15. Ans (B)

$[\lambda g] = LLT^{-2} = L^2T^{-2} = [V^2]$

16. Ans (C)

Blocks A and C both move due to friction. But less friction is available to A as compared to C because normal reaction between A and B is less. Maximum friction between A and B can be:

$$f_{max} = \mu m_A g = \left(\frac{1}{2}\right) mg$$

$\therefore$ Maximum acceleration of A can be

$$a_{max} = \left(\frac{f_{max}}{2}\right) = \frac{g}{2}$$

further $a_{max} = \frac{m_D g}{3m + m_D}$

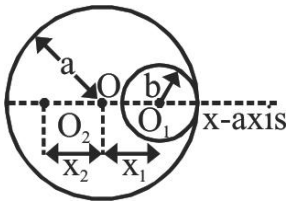
or $\frac{g}{2} = \frac{m_D g}{3m + m_D}$

$\therefore$ (C) is the right answer

18. Ans (A)

The situation can be shown as:

Let radius of complete disc is a and that of small disc is b also let centre of mass now shifts to O_2 at a distance x_2 from original centre.



The position of new cnetre of mass is given by

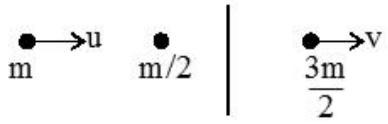
$$X_{CM} = \frac{\sigma.\pi b^2.x_1}{\sigma.\pi a^2 - \sigma.\pi b^2}$$

Here, a = 6cm, b = 2cm, $x_1 = 3.2$ cm

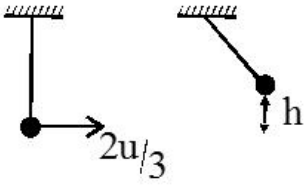
Hence, $X_{CM} = \frac{-\sigma \times \pi(2)^2 \times 3.2}{\sigma \times \pi \times (6)^2 - \sigma \times \pi \times (2)^2}$

$$= \frac{12.8\pi}{32\pi} = 0.4\text{cm}.$$

19. Ans (C)



$mu = \frac{3m}{2}v \Rightarrow v = \frac{2u}{3}$



$\frac{1}{2}\frac{3m}{2}(2u/3)^2 = \frac{3m}{2}gh$

$h = \frac{1}{2}\frac{4u^2}{9g} = \frac{2u^2}{9g}$

20. Ans (B)

$K\varepsilon_{Rot} = \frac{1}{2}\frac{mR^2}{2}.\frac{v^2}{R^2} = \frac{1}{4}mv^2$

$K\varepsilon_{trans} = \frac{1}{2}mv^2$

$K\varepsilon_{Total} = \frac{3}{4}mv^2$

$\frac{K\varepsilon_{Rot}}{K\varepsilon_{Total}} = \frac{1}{3}$

PART-1 : PHYSICS

SECTION-II

1. Ans (650)

$\frac{P}{Q} = \frac{S}{625}$

$\frac{Q}{P} = \frac{S}{676}$

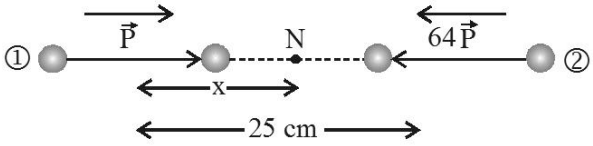
$\therefore \frac{S}{625} = \frac{676}{S}$

$S = 25 \times 26$

$= 650\ \Omega$

2. Ans (5)

Suppose neutral point N lies at a distance x from dipole of moment p or at a distance (25-x) from dipole of 64 p.



At N |E.F. due to dipole ① | = |E.F. due to dipole ②|

$\Rightarrow \frac{1}{4\pi\varepsilon_0}.\frac{2p}{x^3} = \frac{1}{4\pi\varepsilon_0}.\frac{2(64p)}{(25-x)^3}$

$\Rightarrow \frac{1}{x^3} = \frac{64}{(25-x)^3}$

$\Rightarrow$ x = 5 cm

3. Ans (6)

Initially ice will absorb heat to raise it's temperature to 0°C then it's melting takes place

If m_i = Initial mass of ice, m_i' = Mass of ice that melts and m_w = Initial mass of water

By law of mixture, Heat gained by ice = Heat lost by water

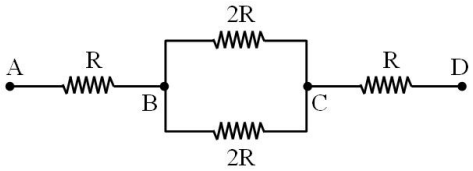
$\Rightarrow m_i \times c \times (20) + m_i' \times L = m_W c_W[20]$

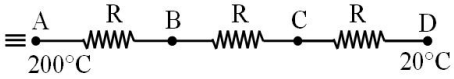
$\Rightarrow 2 \times 0.5(20) + m_i' \times 80 = 5 \times 1 \times 20$

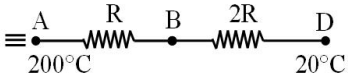
$\Rightarrow m_i' = 1kg$

So final mass of water = Initial mass of water + Mass of ice that melts = 5 + 1 = 6 kg

4. Ans (140)







$\frac{200-T}{R} = \frac{T-20}{2R}$

$400 - 2T = T - 20$

$T = 140°C$

5. Ans (15)

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

$$O = \frac{1}{20} + \frac{1}{f} - \frac{5}{20f}$$

$$O = \frac{20 + f - 5}{20f}$$

$f = -15$ cm

6. Ans (3)

$$\nu_1 = \frac{330}{2 \times l_1}$$

$$\Rightarrow \ell_1 = \frac{330}{2 \times 500} = 33 \text{ cm}$$

Similarly $\ell_2 = 2 \times 33 = 66$ cm

$\ell_3 = 3 \times 33 = 99$cm

$\ell_4 = 4 \times 33 = 132$cm

$\therefore$ Number of resonances will be 3.

7. Ans (3)

$$eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \quad ...(i)$$

$$\frac{eV}{4} = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0} \quad ...(ii)$$

From equation (i) and (ii)

$$\Rightarrow 4 = \frac{\frac{1}{\lambda} - \frac{1}{\lambda_0}}{\frac{1}{2\lambda} - \frac{1}{\lambda_0}}$$ On solving $\lambda_0 = 3\lambda$

8. Ans (44)

$$\frac{6}{0.125} = R + \frac{1}{\left(\frac{1}{8} + \frac{1}{16} + \frac{1}{16}\right)}$$

10. Ans (1)

Angular frequency of pendulum

$$\omega = \sqrt{\frac{g_{eff}}{\ell}}$$

$$\therefore \frac{\Delta\omega}{\omega} = \frac{1}{2}\frac{\Delta g_{eff}}{g_{eff}}$$

$$\Delta\omega = \frac{1}{2}\frac{\Delta g}{g} \times \omega$$

[ω_s = angular frequency of support]

$$\Delta\omega = \frac{1}{2} \times \frac{2A\omega_s^2}{100} \times 100$$

$\Delta\omega = 10^{-3}$ rad/sec.

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)

C N O F
P S Cl
Br
I

In covalent hydrides, acidic strength increase in both directions.

2. Ans (C)

Due to inert pair effect, lower oxidation state is more stable for the heavier members in group 14.

3. Ans (C)

Zn has no unpaired e^-

4. Ans (B)

$K_2Cr_2O_7 + H_2S$

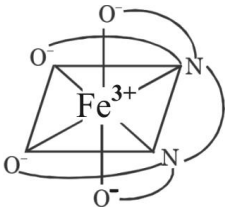
$\longrightarrow Cr_2(SO_4)_3 + S + K_2SO_4 + H_2O$

5. Ans (D)

$[Co(NH_3)_6]^{+3}$ $[CoF_6]^{-3}$

due to SFL, unpaired $e^- = 0$ due to WFL, unpaired $e^- = 4$

6. Ans (C)



No plane of symmetry so optically active. In rest plane of symmetry is present so optically inactive.

7. Ans (D)

% ionic character

$$= \frac{1.2 \times 3.335 \times 10^{-30}}{1.6 \times 10^{-19} \times 10^{-10}} \times 100 = 25\%$$

8. Ans (A)

$K_3 = K_1 \cdot K_2$

$$= \frac{(K_1)^1}{(K_2)^{-1}} \Rightarrow \begin{matrix} x = 1 \\ y = -1 \end{matrix}$$

$\Rightarrow (x - y) = 1 + 1 = 2$

9. Ans (B)

Rate constant $K_{300} = \frac{\ln 2}{40} \text{min}^{-1}$

Rate constant $K_{310} = \frac{\ln 2}{20} \text{min}^{-1}$

$$\ln\left(\frac{K_{310}}{K_{300}}\right) = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$= \frac{E_a}{R}\left(\frac{1}{300} - \frac{1}{310}\right)$$

$$\ln\left(\frac{40}{20}\right) = \frac{E_a}{R}\left(\frac{10}{310 \times 300}\right)$$

$E_a = (\ln 2 \times 2 \times 310 \times 30)$ cal/mol

$= 13020$ cal/mol

10. Ans (B)

$$8 \times V = 2000 \times \frac{20}{100} \times \frac{1}{40}$$

$$V = \frac{10}{8} = 1.25\ \ell$$

$\therefore$ Amount of water added $= 0.75\ \ell$

11. Ans (C)

$BaSO_4 = BaCl_2 + H_2SO_4 - 2HCl$

$\therefore \Lambda^o_{eq}(BaSO_4)$

$= \Lambda^o_{eq}(BaCl_2) + \Lambda^o_{eq}(H_2SO_4) - \Lambda^o_{eq}(HCl)$

$= x_1 + x_2 - x_3$

Now, $\Lambda^o_{eq} = \Lambda_{eq}$ for saturated $BaSO_4$ solution

$$= \frac{1000 \times \kappa}{N}$$

$$\therefore \quad x_1 + x_2 - x_3 = \frac{1000 \times y}{N}$$

$$\Rightarrow N = \frac{10^3 \times y}{x_1 + x_2 - x_3}$$

$$\therefore \quad M = \frac{N}{\text{v. f.}} = \frac{10^3 \times y}{2(x_1 + x_2 - x_3)} = \text{solubility(s)}$$

$$\therefore \quad K_{SP} = S^2 = \frac{10^6 y^2}{4(x_1 + x_2 - x_3)^2}$$

12. Ans (B)

4 gm of NaOH in 100 L sol. $\Rightarrow 10^{-3}$ M sol.

9.8 gm of H_2SO_4 in 100 L sol. $\Rightarrow 10^{-3}$ M sol.

Mixture : 40L of 10^{-3} M NaOH and 10 L of 10^{-3} M H_2SO_4 sol.

Final Conc. of OH^-

$$= \frac{10^{-3}(40 \times 1 - 10 \times 1 \times 2)}{40 + 10} = 6 \times 10^{-4}\text{M}$$

$pOH = -\log(6 \times 10^{-4})$

$= 4 - \log 6 = 4 - 0.60 = 3.40$

$pH = 14 - 3.40 = 10.60$

13. Ans (D)

$$\frac{1.055}{24 + 254 + 18x} \times x \times 18 = (1.055 - 0.695)$$

$\Rightarrow x = 8$ Ans.

14. Ans (A)

$nC_v(T_2 - T_1) = -75$ cal

$$0.1 \times \frac{3R}{2}(T_2 - T_1) = -75$$

$$T_2 - T_1 = \frac{-75}{0.3}$$

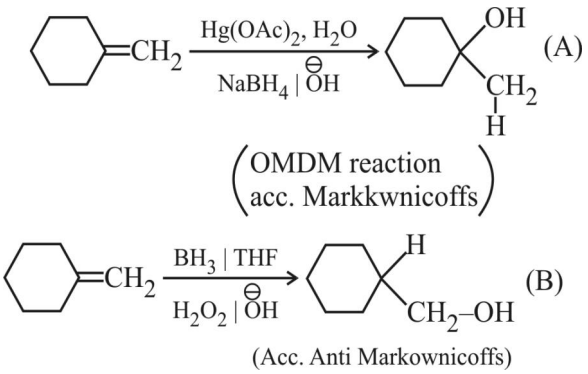
$T_2 - 500 = -250$

$T_2 = 250$ K.

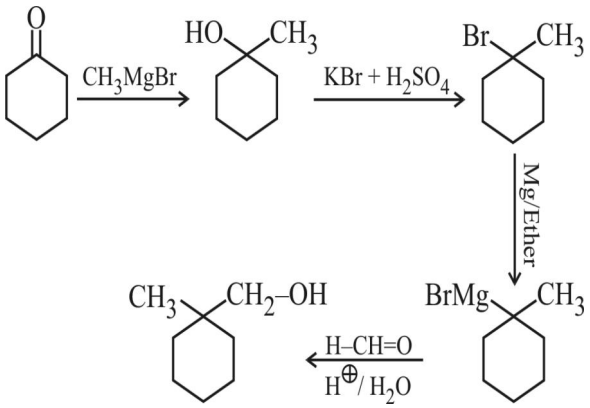
15. **Ans (A)**

Arromatic and extended resonance.

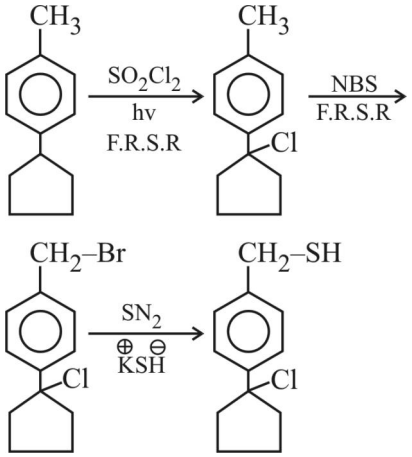
16. **Ans (C)**



18. **Ans (B)**



19. **Ans (B)**



20. **Ans (B)**

Due to presence of EWG $\left[\begin{matrix}O\\ \|\\ R-C-\end{matrix}\right]$ Rate of SN^2 reaction is higher than SN^1 reaction.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (0)**

$K[Ar]4s^1$

$n = 4,$

$\ell = 0,$

$m = 0$

$n \times \ell \times m = 0$

2. **Ans (5)**

5.0

SO_3,PCl_5,CCl_4,PCl_3F_2 and XeF_4 are non-polar

3. **Ans (4)**

Correct Order are :

Order of EN : Zn < Cd < Hg

Order of IE : B > Tl > Ga > Al > In

Order of EA : S > Se > Te > O

Order of atomic radius : Ni < Cu < Zn

4. **Ans (12)**

$t_{1/2}$ = 1 hour = 60 min

$$K = \frac{0.693}{60} = \frac{2.303}{t} \times \log\left(\frac{1}{7/8}\right)$$

$$\frac{2.303 \times \log 2}{60} = \frac{2.303}{t} \times (\log 8 - \log 7)$$

$$t = \frac{60 \times (0.9 - 0.84)}{0.3} = 12 \text{ min}$$

5. **Ans (10)**

$$\alpha = \frac{\Lambda_m}{\Lambda_M^\infty} = \frac{\Lambda_{eq}}{\Lambda_{eq}^\infty}$$

$$\Lambda_{eq} = \frac{K \times 1000}{N}$$

6. Ans (18)

$Cr(OH)_3(s) \rightleftharpoons Cr^{3+}(aq.) + 3OH^{-}(aq.)$

(s) (3s)

$k_{sp} = 27(s)^4 = 6 \times 10^{-31}$

$\Rightarrow [3(s)]^4 = 18 \times 10^{-31}$

$[OH-] = 3(s) = [18 \times 10^{-31}]^{1/4}$

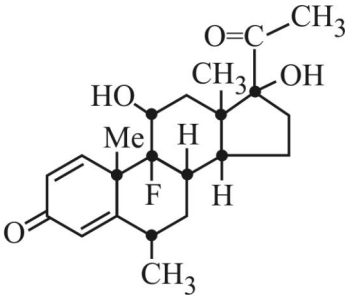
7. Ans (10)

$$\frac{-\Delta P}{P^0} = \frac{n_2}{n_1 + n_2}$$

$$\Rightarrow 0.2 \times 1 = 0.8 \times \frac{x}{40}$$

$\Rightarrow x = 10g$

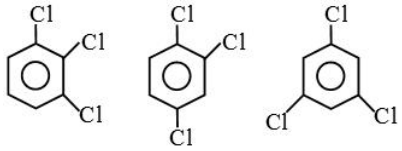
8. Ans (8)



9. Ans (5)

a, c, d, e, h

10. Ans (4)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (A)

$$I = \int \frac{x^8 - x^2}{(x^{12} + 3x^6 + 1)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)} dx$$

Let $\tan^{-1}\left(x^3 + \frac{1}{x^3}\right) = t$

$$\Rightarrow \frac{1}{1 + \left(x^3 + \frac{1}{x^3}\right)^2} \cdot \left(3x^2 - \frac{3}{x^4}\right) dx = dt$$

$$\Rightarrow \frac{x^6}{x^{12} + 3x^6 + 1} \cdot \frac{3x^6 - 3}{x^4} dx = dt$$

$$I = \frac{1}{3}\int \frac{dt}{t} = \frac{1}{3}\ln|t| + C$$

$$I = \frac{1}{3}\ln\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right| + C$$

$$I = \ln\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|^{1/3} + C$$

Hence option (A) is correct

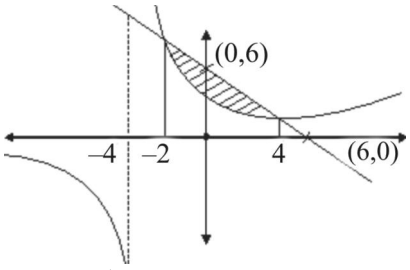
2. Ans (C)

$xy + 4y = 16$, $x + y = 6$

$y(x + 4) = 16$ _(1) , $x + y = 6$__(2)

on solving, (1) & (2)

we get $x = 4, x = -2$



$$\text{Area} = \int_{-2}^{4} \left((6 - x) - \left(\frac{16}{x + 4}\right)\right) dx$$

$= 30 - 32\ln 2$

3. Ans (A)

$\left|P^{-1}AP - 2I\right| = \left|P^{-1}AP - 2P^{-1}P\right|$

$= \left|P^{-1}(A - 2I)P\right|$

$= \left|P^{-1}\right||A - 2I||P|$

$= |A-2I|$

$$= \begin{vmatrix} 0 & 1 & 2 \\ 6 & 0 & 11 \\ 3 & 3 & 0 \end{vmatrix} = 69$$

So, Prime factor of 69 is 3 & 23

So, sum = 26

4. Ans (B)

Given set = {1, 2, 3, ……. 50}

P(A) = Probability that number is multiple of 4

P(B) = Probability that number is multiple of 6

P(C) = Probability that number is multiple of 7

Now, $P(A) = \frac{12}{50}$, $P(B) = \frac{8}{50}$, $P(C) = \frac{7}{50}$

again $P(A \cap B) = \frac{4}{50}, P(B \cap C)$

$= \frac{1}{50}, P(A \cap C) = \frac{1}{50}$

$P(A \cap B \cap C) = 0$

Thus $P(A \cup B \cup C)$

$= \frac{12}{50} + \frac{8}{50} + \frac{7}{50} - \frac{4}{50} - \frac{1}{50} - \frac{1}{50} + 0 = \frac{21}{50}$

5. Ans (D)

$$D = \begin{vmatrix} 1 & -2 & 1 \\ 2 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix}$$

$= 1(ab + 3) + 2(2b - 9) + 1(-2 - 3a)$

$= ab + 3 + 4b - 18 - 2 - 3a$

For infinite solutions $D = 0$, $D_1 = 0$, $D_2 = 0$ and $D_3 = 0$

$D = 0$

$ab - 3a + 4b = 17$ ….(1)

$$D_1 = \begin{vmatrix} -4 & -2 & 1 \\ 5 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix} = 0$$

$$D_2 = \begin{vmatrix} 1 & -4 & 1 \\ 2 & 5 & 3 \\ 3 & 3 & \beta \end{vmatrix} = 0$$

$\Rightarrow 1(5\beta - 9) + 4(2\beta - 9) + 1(6 - 15) = 0$

$13\beta - 9 - 36 - 9 = 0$

$13\beta = 54, \beta = \frac{54}{13}$ put in (1)

$\frac{54}{13}\alpha - 3\alpha + 4\left(\frac{54}{13}\right) = 17$

$54\alpha - 39\alpha + 216 = 221$

$15\alpha = 5 \quad \alpha = \frac{1}{3}$

Now, $12\alpha + 13\beta = 12.\frac{1}{3} + 13.\frac{54}{13}$

$= 4 + 54 = 58$

6. Ans (A)

$\log_e a, \log_e b, \log_e c$ are in A.P.

$\therefore b^2 = ac$ …..(i)

Also

$\log_e\left(\frac{a}{2b}\right), \log_e\left(\frac{2b}{3c}\right), \log_e\left(\frac{3c}{a}\right)$

are in A.P. $\left(\frac{2b}{3c}\right)^2 = \frac{a}{2b} \times \frac{3c}{a}$

$\frac{b}{c} = \frac{3}{2}$

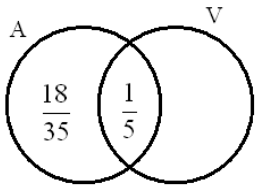
Putting in eq. (i)

$b^2 = a \times \frac{2b}{3}$

$\frac{a}{b} = \frac{3}{2}$

$a : b : c = 9 : 6 : 4$

7. Ans (B)



$P\left(\bar{A}\right) = \frac{2}{7} = p$

$P(A \cap V) = \frac{1}{5} = q$

$P(A) = \frac{5}{7}$

$P\left(A \cap \bar{V}\right) = \frac{18}{35}$

8. Ans (B)

$$A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix} \Rightarrow \det(A) = 3$$

$$B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow \det(B) = 1$$

Now $C = ABA^T \Rightarrow \det(C) = (\det(A))^2 \times \det(B)$

Now $|X| = |A^T C^2 A|$

$= |A^T| \, |C|^2 \, |A|$

$= |A|^2 \, |C|^2$

$= 9 \times 81$

$= 729$

9. Ans (B)

$x^2 - x - 1 = 0$

$S_n = 2023\alpha^n + 2024\beta^n$

$S_{n-1} + S_{n-2} = 2023\alpha^{n-1} + 2024\beta^{n-1}$

$+2023\alpha^{n-2} + 2024\beta^{n-2}$

$= 2023\alpha^{n-2}[1 + \alpha] + 2024\beta^{n-2}[1 + \beta]$

$= 2023\alpha^{n-2}[\alpha^2] + 2024\beta^{n-2}[\beta^2]$

$= 2023\alpha^n + 2024\beta^n$

$S_{n-1} + S_{n-2} = S_n$

Put n = 12

$S_{11} + S_{10} = S_{12}$

10. Ans (B)

$\begin{vmatrix} \lambda & 3 \\ 2 & 6 \end{vmatrix} > 0 \Rightarrow 6\lambda - 6 > 0$

11. Ans (C)

Hence f(x) is continuous at $x = \frac{\pi}{2}$

$\Rightarrow$

$$K = \lim_{x \to \frac{\pi}{2}} \frac{1 - \sin x}{(\pi - 2x)^2} \cdot \frac{\log \sin x}{\log(1 + \pi^2 - 4\pi x + 4x^2)}$$

Put $\alpha = \left(\frac{\pi}{2} - x\right)$

$$K = \lim_{\alpha \to 0} \frac{1 - \cos x}{(2\alpha)^2} \cdot \frac{\log \cos \alpha}{\log(1 + 4\alpha^2)}$$

$$= \lim_{\alpha \to 0} (1 - \cos \alpha) \frac{\log(1 + \cos \alpha - 1)}{(\cos \alpha - 1)}$$

$$\frac{4\alpha^2}{\log(1 + 4\alpha^2)} \frac{(\cos \alpha - 1)}{(2\alpha)^2 4\alpha^2}$$

$$= \lim_{\alpha \to 0} \frac{1 - \cos \alpha}{\left(\frac{\alpha^2}{4}\right)} \frac{(\cos \alpha - 1)}{\frac{\alpha^2}{4}} \frac{\frac{\alpha^2}{4} \cdot \frac{\alpha^2}{4}}{4\alpha^2 . 4\alpha^2}$$

$$K = \frac{2.(-2)}{4.4.4.4} = -\frac{1}{64}$$

12. Ans (A)

Let $\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8} = \theta$

$\sin 4\theta = \frac{\sqrt{63}}{8}$

$\cos 4\theta = \frac{1}{8}$

$2\cos^2 2\theta - 1 = \frac{1}{8}$

$\cos^2 2\theta = \frac{9}{16}$

$\cos 2\theta = \frac{3}{4}$

$2\cos^2 \theta - 1 = \frac{3}{4}$

$\cos^2 \theta = \frac{7}{8}$

$\cos \theta = \frac{\sqrt{7}}{2\sqrt{2}}$

$\tan \theta = \frac{1}{\sqrt{7}}$

13. Ans (A)

$$\sqrt{1 + \cos x} = \sqrt{2\cos^2 \frac{x}{2}} = \sqrt{2}\left|\cos \frac{x}{2}\right|$$

$$\sqrt{1 - \cos x} = \sqrt{2}\left|\sin \frac{x}{2}\right|$$

$$\therefore \frac{\sqrt{1 + \cos x} + \sqrt{1 - \cos x}}{\sqrt{1 + \cos x} - \sqrt{1 - \cos x}} = \frac{\left|\cos \frac{x}{2}\right| + \left|\sin \frac{x}{2}\right|}{\left|\cos \frac{x}{2}\right| - \left|\sin \frac{x}{2}\right|}$$

$$= \frac{-\cos \frac{x}{2} + \sin \frac{x}{2}}{-\cos \frac{x}{2} - \sin \frac{x}{2}} = \frac{1 - \tan \frac{x}{2}}{1 + \tan \frac{x}{2}} = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)$$

$$= \cot\left[\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{x}{2}\right)\right] = \cot\left(\frac{\pi}{4} + \frac{x}{2}\right)$$

$$\therefore \boxed{a = \frac{\pi}{4}}$$

14. Ans (C)

$$s = \sqrt{\frac{1}{n}\Sigma d^2 - \left(\frac{\Sigma d}{n}\right)^2}$$

$$\sqrt{\frac{1}{5}(25) - \left(\frac{5}{5}\right)^2} = 2$$

S.D. if affected with change in scale so S.D. of $2x; +7$ if 4

15. Ans (A)

$$P = 96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$$

$$2P \times \sin\frac{\pi}{33} = 96 \times 2\sin\frac{\pi}{33}\cos\frac{\pi}{33}$$

$$\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$$

$$2P \times \sin\frac{\pi}{33} = 6 \times \sin\frac{32\pi}{33} = 6\sin\frac{\pi}{33}$$

$P = 3$

16. Ans (C)

Let $\vec{a} = \lambda\vec{b} + \mu\vec{c}$

is equally inclined to $\vec{b}$ and $\vec{d}$, where $\vec{d} = \hat{j} + 2\hat{k}$.

$$\Rightarrow \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{\vec{a}\cdot\vec{d}}{|\vec{a}||\vec{d}|}$$

$$\Rightarrow \frac{\left(\lambda\vec{b} + \mu\vec{c}\right).\vec{b}}{\sqrt{5}} = \frac{\left(\lambda\vec{b} + \mu\vec{c}\right)\cdot\vec{d}}{\sqrt{5}}$$

$$\frac{\left[\lambda\left(2\hat{i} + \hat{j}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)\right]\cdot\left(2\hat{i} + \hat{j}\right)}{\sqrt{5}}$$

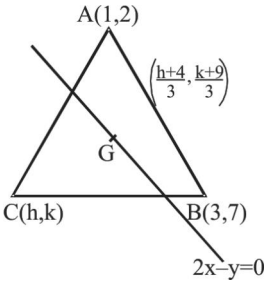
$$= \frac{\left[\lambda\left(2\hat{i} + \hat{j}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)\right]\cdot\left(\hat{j} + 2\hat{k}\right)}{\sqrt{5}}$$

or $\lambda(4 + 1) + \mu(2-1) = \lambda\ (1) + \mu(-1+2)$

or $4\lambda = 0$, i.e., $\lambda = 0$

$$\therefore \hat{a} = \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

17. Ans (D)



Let vertex C is (h, k)

$$\therefore \text{Centroid} = \left(\frac{h+4}{3}, \frac{k+9}{3}\right)$$

Centroid lies on the line $2x - y = 0$

$$\therefore \frac{2h+8}{3} = \frac{k+9}{3}$$

$\Rightarrow$ Locus of C is $2x - y = 1$

$$\therefore \text{Minimum distance} = \frac{1}{\sqrt{5}}$$

20. Ans (A)

From the first relation, $n = -\left(-\frac{a\ell + bm}{c}\right)$

Put the value of n in second relation,

$$fm\left(-\frac{(a\ell + bm)}{c}\right) + g\ell\left(-\frac{(a\ell + bm)}{c}\right) + h\ell m = 0$$

or $afm\ell + bfm^2 + ag\ell^2 + bg\ell m - ch\ell m = 0$

$$ag\frac{\ell^2}{m^2} + \frac{\ell}{m}(af + bg - ch) + bf = 0 \qquad ....(i)$$

Now if ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 be direction cosines of two lines, then from (i)

$$\frac{\ell_1\ell_2}{m_1m_2} = \frac{bf}{ag}, \left[\text{Since roots of (i) are} \frac{\ell_1}{m_1}\ \frac{\ell_2}{m_2}\right]$$

or $\frac{\ell_1\ell_2}{f/a} = \frac{m_1m_2}{g/b} = \frac{n_1n_2}{h/c} = q$ (say)

We know that the lines are perpnedicular, if

$\ell_1\ell_2 + m_1m_2 + n_1n_2 = 0$

i.e., $\left(\frac{f}{a}\right)q + \left(\frac{g}{b}\right)q + \left(\frac{h}{c}\right)q = 0$ or

$$\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$$

Note : Student should remember this question as a fact.

PART-3 : MATHEMATICS

SECTION-II

1. Ans (11)

$$\frac{dy}{dx} = \frac{x+y-2}{x-y}$$

$x = X + h, y = Y + k$

$$\frac{dY}{dX} = \frac{X+Y}{X-Y}$$

$$\left.\begin{matrix} h+k-2=0 \\ h-k=0 \end{matrix}\right\} h = k = 1$$

$Y = vX$

$$v + \frac{dv}{dX} = \frac{1+v}{1-v}$$

$$\Rightarrow X - \frac{dv}{dX} = \frac{1+v^2}{1-v}$$

$$\frac{1-v}{1+v^2}dv = \frac{dX}{X}$$

$$\tan^{-1}v - \frac{1}{2}\ln(1+v^2) = \ln|X| + C$$

As curve is passing through (2, 1)

$$\tan^{-1}\left(\frac{y-1}{x-1}\right) - \frac{1}{2}\ln\left(1+\left(\frac{y-1}{x-1}\right)^2\right) = \ln|x-1|$$

$\therefore \alpha = 1$ and $\beta = 2$

$\Rightarrow 5\beta + \alpha = 11$

2. Ans (14)

$(t+1)dx = (2x + (t+1)^4)dt$

$$\frac{dx}{dt} = \frac{2x + (t+1)^4}{t+1}$$

$$\frac{dx}{dt} - \frac{2x}{t+1} = (t+1)^3$$

$$I \times F = e^{-\int \frac{2}{t+1}dt} = e^{-2\ln(t+1)} = \frac{1}{(t+1)^2}$$

$$\frac{x}{(t+1)^2} = \int \frac{1}{(t+1)^2}(t+1)^3 dt + c$$

$$\frac{x}{(t+1)^2} = \frac{(t+1)^2}{2} + c$$

$$\Rightarrow c = \frac{3}{2}$$

$$x = \frac{(t+1)^4}{2} + \frac{3}{2}(t+1)^2$$

put, $t = 1$

$x = 2^3 + 6 = 14$

3. Ans (5)

$x^2 + x + 1 = 0 \Rightarrow x = \omega, \omega^2 = \alpha$

Let $a = \omega$

Now $(1+\alpha)^7 = -\omega^{14} = -\omega^2 = 1 + \omega$

$A = 1, B = 1, C = 0$

$\therefore 5(3A - 2B - C) = 5(3 - 2 - 0) = 5$

4. Ans (118)

$$(1+x)(1-x^2)\left(1 + \frac{3}{x} + \frac{3}{x^2} + \frac{1}{x^3}\right)^5$$

$$= (1+x)(1-x^2)\left(\left(1+\frac{1}{x}\right)^3\right)^5$$

$$= \frac{(1+x)^2(1-x)(1+x)^{15}}{x^{15}}$$

$$= \frac{(1+x)^{17} - x(1+x)^{17}}{x^{15}}$$

$= \text{coeff}\left(x^3\right)$ in the expansion

$\approx \text{coeff}\left(x^{18}\right)$ in $(1+x)^{17} - x(1+x)^{17}$

$$= \binom{17}{2} - \binom{17}{1}$$

$= 17 \times 8 - 17$

$= 17 \times 7$

$= 119$

Hence Answer $= 119 - 1 = 118$

5. Ans (1)

$F'(x) = e^{-x^2/2}(1 - x^2) = 0$

$\therefore x = \pm 1$

7. Ans (4)

$$\sin^2\alpha = \frac{-1+\sqrt{5}}{2} \Rightarrow \cos 2\alpha = 2 - \sqrt{5}$$

$(\cos^8\alpha - \sin^8\alpha) = (\cos^4\alpha + \sin^4\alpha)\cos^2\alpha$

$$= \left(1 - \frac{1}{2}\sin^2 2\alpha\right)\cos 2\alpha = 20 - 9\sqrt{5}$$

8. Ans (5)

The smallest equivalence relation is the identify relation

$R_1 = \{(a, a), (b, b), (c, c)\}$

Then, two ordered pairs of two distinct elements can be added to give three more equivalence relations

$R_2 = \{(a, a), (b, b), (c, c), (a, b) (b, a)\}$

Similarly R_3 and R_4.

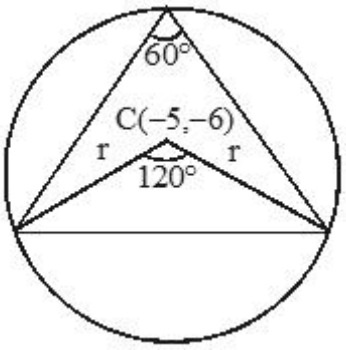
Finally, the largest equivalence relation, that is the universal

$R_5 = \{(a, a), (b, b), (c, c), (a, b),(b, a),(a, c),(c, a), (b, c), (c, b)\}$

10. Ans (25)

$$3\left(\frac{1}{2}r^2.\sin 120^\circ\right) = 27\sqrt{3}$$

$$\frac{r^2}{2}\frac{\sqrt{3}}{2} = \frac{27\sqrt{3}}{3}$$



$$r^2 = \frac{108}{3} = 36$$

Radius $= \sqrt{25 + 36 - C} = \sqrt{36}$

$\boxed{C = 25}$